

Research Methodology in CS (CSEG3060_3)

Lecture 6: International Cooperation on Intellectual Property and Procedures for Patent Grants: Mechanisms of international cooperation (WIPO, bilateral and regional agreements), cross-border filing procedures, and procedural aspects of patent grants across jurisdictions.

Unit 5: Intellectual Property and Patents • Mapped Course Outcome: CO3

Lecture 6: International Cooperation on Intellectual Property and Procedures for Patent Grants

Course: Research Methodology in Computer Science (CSEG3060_3)

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Lecture Overview

This lecture examines the architecture of international cooperation in intellectual property (IP) governance, focusing on how researchers, inventors, and technology companies secure patent protection across multiple jurisdictions. As computer science innovations increasingly transcend national boundaries—through software, algorithms, semiconductor designs, and artificial intelligence systems—understanding global IP mechanisms becomes essential for conducting responsible, legally sound, and commercially viable research.

The lecture is structured around three interconnected themes: (i) institutional and treaty-based mechanisms that facilitate international IP cooperation, (ii) procedural pathways for cross-border patent filing, and (iii) substantive and procedural variations in patent grant examinations across major jurisdictions.

Learning Objectives

Upon successful completion of this lecture, students will be able to:

1. **Describe** the principal institutional frameworks for international cooperation on intellectual property, including the role of the World Intellectual Property Organization (WIPO) and key multilateral treaties.
2. **Differentiate** between bilateral, regional, and multilateral IP cooperation mechanisms and evaluate their relevance to computer science research outputs.
3. **Analyze** cross-border patent filing procedures, including the Paris Convention priority system and the Patent Cooperation Treaty (PCT) international phase.
4. **Compare** procedural and substantive aspects of patent grant examinations across the United States, Europe, Japan, China, and India.
5. **Formulate** strategic filing recommendations for software-related inventions, AI algorithms, and other computer science innovations in an international context.
6. **Apply** treaty-based procedures to hypothetical research scenarios involving multinational patent protection.

1. Introduction: The Imperative of International IP Cooperation

Innovation in computer science rarely remains confined to a single jurisdiction. A machine learning model developed in Bangalore may be deployed on servers in Frankfurt and accessed from São Paulo. A cryptographic protocol invented in California may be implemented across global networks. Under the territoriality principle, patent protection granted in one country does not automatically extend to another. This creates a fundamental challenge: inventors must seek protection in each country where they desire exclusive rights.

Historically, this required duplicate filings in every country, with parallel translations, multiple attorneys, and varied procedural requirements. The twentieth and twenty-first centuries witnessed the development of international cooperation frameworks that reduce redundancy, harmonize substantive standards, and streamline administrative burdens. The lecture traces the evolution of these frameworks from the Paris Convention (1883) to contemporary regional and bilateral arrangements.

The computer science researcher must understand these mechanisms not merely as legal formalities but as integral components of research strategy—affecting decisions about publication timing, technology transfer agreements, open-source licensing, and commercialization pathways.

2. Key Concepts and Definitions

Term	Definition
Territoriality Principle	Patent rights are limited to the geographic boundaries of the issuing sovereign state.
Priority Right	The right granted to the first patent applicant to claim precedence for subsequent filings in other jurisdictions within a stipulated period.
International Cooperation	Formal arrangements between states to harmonize, streamline, or mutually recognize aspects of IP protection.
Multilateral Treaty	An agreement among three or more sovereign states governing reciprocal rights and obligations.
Regional Agreement	A treaty among a defined group of states (typically contiguous or economically integrated) that establishes a common IP regime.
Bilateral Agreement	An agreement between two states addressing specific aspects of IP cooperation.
PCT (Patent Cooperation Treaty)	A multilateral treaty facilitating unified international patent filing procedures administered by WIPO.
National Phase	The stage at which a PCT application is converted into individual national or regional patent applications.
Substantive Examination	An in-depth assessment of patentability criteria (novelty, inventive step, industrial applicability).
Grace Period	A defined interval after public disclosure during which an inventor may still file a patent application without losing novelty.

3. Mechanisms of International Cooperation

3.1 The World Intellectual Property Organization (WIPO)

The World Intellectual Property Organization (WIPO) is a specialized agency of the United Nations headquartered in Geneva, Switzerland. Established by the WIPO Convention of 1967, WIPO became a UN agency in 1974. With 193 member states, WIPO administers 26 international treaties and provides a global forum for IP policy development.

3.1.1 Core Functions of WIPO

- 1. Treaty Administration:** WIPO administers major treaties including the Paris Convention, the Patent Cooperation Treaty (PCT), the Madrid System for trademarks, the Hague System for industrial designs, and the Budapest Treaty for microorganism deposits.
- 2. International Registration Systems:** The PCT System, the Madrid System for the International Registration of Marks, and the Hague System for the International Registration of Industrial Designs operate under WIPO administration.
- 3. Policy Coordination:** WIPO facilitates intergovernmental negotiations on emerging IP issues, including genetic resources, traditional knowledge, and artificial intelligence.
- 4. Technical Assistance:** WIPO provides capacity-building programs, particularly for developing countries, including training, databases (such as PATENTSCOPE), and classification systems (International Patent Classification, IPC).
- 5. Dispute Resolution:** WIPO offers arbitration and mediation services for IP disputes, including domain name disputes through the WIPO Arbitration and Mediation Center.

3.1.2 Significance for Computer Science Research

For computer science researchers, WIPO offers several practical benefits:

- A **centralized international filing system** through the PCT reduces administrative complexity.
- The **PATENTSCOPE database** provides free access to over 100 million patent documents, facilitating prior art searches essential for novelty assessment.
- **WIPO standards** for documentation, classification, and search reports enhance the quality and comparability of patent information globally.
- **WIPO IP Policy and Technology Trends reports**, such as the influential 2019 *Technology Trends: Artificial Intelligence* report, inform policy debates on AI-related IP.

3.2 Multilateral Treaties

3.2.1 The Paris Convention for the Protection of Industrial Property (1883)

The Paris Convention is the foundational treaty of international patent cooperation. Although predating WIPO, it is administered by WIPO and remains central to cross-border filing.

Key Provisions:

- **National Treatment (Article 2):** Each member state must grant nationals of other member states the same protection it grants to its own nationals.
- **Right of Priority (Article 4):** An applicant who files a patent application in one member state has a priority period of **12 months** to file in other member states. Subsequent filings within this period are treated as if filed on the original filing date, preserving novelty against intervening disclosures.
- **Independence of Patents (Article 4bis):** Patents obtained in different member states are independent of each other.
- **Common Rules:** Provisions on compulsory licensing, grace periods for exhibition disclosures, and inventor's right to be named.

Worked Example (Computer Science Context):

A researcher at IIT Delhi develops a novel federated learning algorithm and files a provisional patent application in India on 15 March 2025. On 10 February 2026—within the 12-month priority window—she files a corresponding application in the United States. The US application is treated, for novelty purposes, as if filed on 15 March 2025, thereby excluding any prior art published between those dates.

3.2.2 The Patent Cooperation Treaty (PCT) (1970)

The PCT represents the most significant operational mechanism for international patent cooperation. Administered by WIPO, the PCT unifies the procedural initial stages of patent filing across its 150+ contracting states.

The PCT operates in two principal phases:

A. International Phase (approximately 18-30 months from priority date)

1. **International Filing:** A single application is filed with the Receiving Office (the national patent office of a PCT contracting state, a regional office, or WIPO itself), designating the contracting states in which the applicant seeks protection.
2. **International Search:** An International Searching Authority (ISA)—such as the European Patent Office, the Korean Intellectual Property Office, or the Indian Patent Office—identifies prior art and issues an **International Search Report (ISR)** and a **Written Opinion of the International Searching Authority** on patentability.
3. **International Publication:** Approximately 18 months after the priority date, the application is published by WIPO and becomes accessible via PATENTSCOPE.
4. **Preliminary Examination (Optional):** Through the Patent Prosecution Highway (PPH) and the International Preliminary Examining Authority, applicants may obtain a preliminary assessment on patentability. This is particularly valuable for refining claims before national phase entry.

B. National/Regional Phase (from 30 months from priority date, with variations)

Within 30 months (most jurisdictions) or 31/32 months (some), the applicant must:

- File translations into the official language(s) of each designated office.
- Pay national fees.
- Engage local patent attorneys (in many jurisdictions).
- Prosecute the application before the national or regional patent office.

Advantages of the PCT System:

- **Delayed Decision-Making:** The 30-month international phase delays costly national filings, providing time to assess commercial viability and refine claims.
- **Streamlined Procedures:** A single international filing replaces multiple national filings.
- **Preliminary Assessment:** The ISR and Written Opinion provide valuable guidance on patentability.
- **Marketing Tool:** PCT publication may serve to attract licensees or investors.

Limitations of the PCT:

- The PCT does not grant an "international patent." It is a filing and search procedure, not a grant procedure.

- National phase entry remains complex, requiring jurisdiction-specific filings.
- Fees accumulate across international and national phases.

PCT Filing Flow (Schematic Representation):

Priority Date → International Filing → International Search → International Publication → Preliminary Examination (Optional) → National/Regional Phase Entry
 Priority Date → International Filing → International Search → International Publication → Preliminary Examination (Optional) → National/Regional Phase Entry

LaTeX representation of timing:

$$t_{\text{national phase entry}} \approx t_{\text{priority date}} + 30 \text{ months}$$

$t_{\text{national phase entry}} \approx t_{\text{priority date}} + 30 \text{ months}$

3.2.3 Other WIPO-Administered Treaties Relevant to Computer Science

- **The Budapest Treaty on the International Recognition of the Deposit of Microorganisms (1977)**: Relevant where inventions involve biological components (e.g., bioinformatics, biotech-CS convergence).
- **The Strasbourg Agreement Concerning the International Patent Classification (1971)**: Establishes the IPC, a standardized classification system used by patent offices worldwide.
- **The Vienna Agreement Establishing an International Classification of the Figurative Elements of Marks (1973)**: Less directly relevant but part of the IP classification infrastructure.

3.3 Regional Patent Systems

Regional agreements establish unified or harmonized patent regimes among a group of states. The principal examples are as follows:

3.3.1 The European Patent Convention (EPC) and the European Patent Office (EPO)

The EPC, signed in 1973, established the European Patent Office (EPO) headquartered in Munich. It is not an EU instrument (though EU member states are party to it), and post-Brexit the United Kingdom remains within the EPC framework.

Key Features:

- A **single European patent application** may designate multiple contracting states (currently 39).
- The EPO conducts **substantive examination** centrally, including novelty, inventive step, and industrial applicability.
- Upon grant, the European patent **bundles** into nationally enforceable patents, with validation in designated states.
- The **Unified Patent Court (UPC)**, operational from 2023, provides a unified litigation forum for participating EU member states.

Computer Science Significance:

- The EPO has developed extensive jurisprudence on **computer-implemented inventions (CIs)**, including the well-known *COMVIK approach* requiring a "technical character" assessment.
- AI-related patent examination at the EPO has been clarified through recent Guidelines for Examination (2024), providing greater clarity for neural networks, training methods, and machine learning systems.

3.3.2 The African Regional Intellectual Property Organization (ARIPO) and the African Intellectual Property Organization (OAPI)

- **ARIPO** (Headquartered in Harare, Zimbabwe): Operates under the Harare Protocol on Patents and Industrial Designs (1982), covering predominantly English-speaking African states.
- **OAPI** (Headquartered in Yaoundé, Cameroon): Operates under the Bangui Agreement (1977), covering predominantly francophone African states, with a single registration covering all member states.

Both systems are increasingly relevant as African markets expand for software products and digital services.

3.3.3 The Eurasian Patent Convention (EAPC)

A regional system established among former Soviet states, administered by the Eurasian Patent Organization (EAPO) headquartered in Moscow. Members include Russia, Belarus, Kazakhstan, and others.

3.3.4 Patent Law Treaties (PLT) and Substantive Patent Law Treaty (SPLT) Negotiations

The Patent Law Treaty (2000) harmonizes formalities (filing date requirements, representation rules, restoration of priority rights). Negotiations toward a Substantive Patent Law Treaty continue under WIPO auspices, though they remain contentious given divergences in substantive standards.

3.4 Bilateral Agreements

Bilateral agreements address IP cooperation between two states. They may take various forms:

- **Bilateral Patent Prosecution Highway (PPH) Agreements:** Allow accelerated examination in one country based on a positive examination in another. WIPO and individual patent offices have established PPH networks.
- **Bilateral IP Treaties:** Examples include the US-Japan Patent Prosecution Highway Agreement.
- **Free Trade Agreements (FTAs):** Modern FTAs contain substantial IP chapters, often elevating IP standards beyond TRIPS. Examples include the USMCA (US-Mexico-Canada), the EU-Korea FTA, and the Comprehensive and Progressive Agreement for Trans-Pacific Partnership (CPTPP).

Significance for Computer Science:

- **PPH programs** are particularly valuable for software patent applicants, allowing rapid prosecution across multiple jurisdictions when one office has found the invention patentable.
- **FTA IP chapters** may impose obligations regarding software copyright, patent term restoration for regulatory delays, and limitations on compulsory licensing—matters relevant to CS companies.

4. Cross-Border Filing Procedures: A Detailed Procedural Analysis

4.1 Strategic Considerations Before Filing

Before initiating cross-border filing, a CS researcher should evaluate:

1. **Commercial Markets:** Where will the invention be manufactured, sold, or licensed?
2. **Competitor Activity:** Where do competitors file, and where is infringement most likely?
3. **Enforcement Infrastructure:** Which jurisdictions offer effective patent enforcement?
4. **Cost-Benefit Analysis:** Filing costs across jurisdictions must be weighed against anticipated returns.
5. **Publication Strategy:** Premature publication can prejudice patentability (with limited grace periods in jurisdictions like the US and certain others).
6. **Open-Source Considerations:** Public disclosure through GitHub, arXiv, or conference papers affects novelty, requiring careful timing relative to filing.

4.2 The Paris Convention Route vs. The PCT Route

Dimension	Paris Convention Direct Filing	PCT Route
Initial Cost	Lower (no PCT fees)	Higher (PCT fees)
Delay Before National Filings	None (must file within 12 months)	Approximately 18 additional months
Procedural Consolidation	None	Unified international phase
Search Report Available	No (unless requested individually)	Yes (ISR from ISA)
Suitable For	Single-country filings or rapid prosecution	Multi-jurisdiction filings with delayed decision
Total Jurisdictions Typically Pursued	1-2	3+

4.3 Detailed PCT Procedural Steps

The following is a structured procedural overview:

Step 1: Preparation

- Draft patent specification including description, claims, abstract, drawings.
- Identify priority application (if any).
- Determine contracting states to be designated.

Step 2: International Filing

- File Form PCT/RO/101 (or equivalent e-filing via ePCT) with the Receiving Office.
- Pay transmittal, search, and international filing fees.
- Receive international filing date and international application number.

Step 3: International Search

- The ISA conducts prior art search.
- Issues ISR and Written Opinion (typically within 16 months from priority date).

Step 4: International Publication

- WIPO publishes the application approximately 18 months from priority date.
- Application available via PATENTSCOPE.

Step 5: (Optional) Preliminary Examination

- Under Chapter II of the PCT.
- Applicant files demand (Form PCT/IPEA/401).
- IPEA issues International Preliminary Report on Patentability (IPRP).

Step 6: National Phase Entry

- Within 30 months (typically) from priority date.
- File translations, national fees, and attorney representation.
- Prosecution proceeds in each designated jurisdiction.

Mathematical Illustration of Timing:

If priority date is t_0 , then:

$$t_{\text{ISR}} \approx t_0 + 16 \text{ months}$$
$$t_{\text{PCT publication}} \approx t_0 + 18 \text{ months}$$
$$t_{\text{national phase entry}} \approx t_0 + 30 \text{ months}$$

$$t_{\text{ISR}} \approx t_0 + 16 \text{ months}$$
$$t_{\text{PCT publication}} \approx t_0 + 18 \text{ months}$$
$$t_{\text{national phase entry}} \approx t_0 + 30 \text{ months}$$

4.4 Regional Filing Routes

For jurisdictions participating in regional systems, applicants may file:

- **Direct regional application** (e.g., European patent application at the EPO).
- **PCT regional phase entry** (most PCT applicants use this route for European validation).
- **Validation through regional office** (e.g., ARIPO filing designating multiple African states).

5. Procedural Aspects of Patent Grants Across Jurisdictions

5.1 Substantive Patentability Requirements: Comparative Analysis

While the Paris Convention and TRIPS Agreement establish minimum standards, substantive patent law varies significantly across jurisdictions.

5.1.1 Novelty

Jurisdiction	Standard	Reference
United States	"Absolute novelty" with grace period (6 months) for inventor's own disclosures	35 USC §102(a)(b)
Europe (EPC)	Strict novelty; no grace period for technical disclosures (except limited exhibition exception)	EPC Article 54
Japan	Strict novelty; 6-month grace period for inventor's own disclosures	Japan Patent Act
China	Strict novelty (with limited exceptions for academic conferences)	PRC Patent Law
India	Strict novelty; limited grace period for display at exhibitions	Indian Patents Act

Practical Implication for CS Researchers:

Publishing a paper on arXiv before filing in a strict-novelty jurisdiction destroys novelty. Researchers must therefore time publications carefully or file before disclosure.

5.1.2 Inventive Step / Non-Obviousness

- **United States:** "Non-obviousness" under 35 USC §103, assessed from the perspective of a person having ordinary skill in the art (PHOSITA), with secondary considerations (Graham v. John Deere factors).
- **Europe:** "Inventive step" under EPC Article 56, assessed using the "problem-solution approach" established in EPO jurisprudence.
- **Japan:** "Inventive step" requiring that the invention would not be "easily conceivable" by a person skilled in the art.

5.1.3 Industrial Applicability / Utility

- **Europe (EPC):** Requires "industrial application" (Article 57). Pure mathematical methods or abstract algorithms without technical application may fail this requirement.
- **United States:** Requires "useful" invention under 35 USC §101. Recent Supreme Court decisions (*Alice Corp. v. CLS Bank*, 2014; *Mayo v. Prometheus*, 2012) have imposed a two-step Alice/Mayo test for abstract ideas, significantly affecting software patentability.
- **India:** Section 2(1)(ac) and Section 3 of the Patents Act exclude "computer programs per se" from patentability, though hardware-implemented inventions with technical effect may qualify.

5.2 Procedural Variations in Examination

Procedural Element	United States (USPTO)	Europe (EPO)	Japan (JPO)	China (CNIPA)	India (IPO)
Examination Type	Substantive	Substantive	Substantive	Substantive	Substantive
Examination Request	Substantive examination not automatic; must be requested within 3 months of publication	Automatic upon filing (with examination request)	Automatic upon filing	Automatic upon filing	Within 48 months of priority date
Average Pendency	~20-24 months	~24-30 months	~16-18 months	~18-22 months	Variable (often long)
Opposition	Post-grant review, IPR	9-month opposition period after grant	Post-grant opposition	Post-grant invalidation	Post-grant opposition
Software Patent Approach	Alice/Mayo test (restrictive)	Technical character requirement (CII Guidelines)	Generally favorable to software	Increasingly permissive	Restricted under Section 3(k)
Patent Term	20 years from filing	20 years from filing	20 years from filing	20 years from filing	20 years from filing

5.3 Software and AI Patentability Across Jurisdictions

This subsection merits particular attention given its relevance to computer science research.

5.3.1 United States

Under the *Alice/Mayo* two-step framework:

Step 1: Does the claim recite a patent-ineligible concept (law of nature, natural phenomenon, abstract idea)?

Step 2: If so, is there an "inventive concept" sufficient to transform the claim into a patent-eligible application?

Many software and AI-related claims have been invalidated under this framework, particularly pure business method patents and abstract algorithms. However, claims tied to specific technological improvements, technical effects, or novel hardware architectures are more likely to survive.

5.3.2 European Patent Office

Under the *COMVIK approach* (T 641/00) and the EPO's Guidelines for Examination (2024 revision), a computer-implemented invention is examined for "technical character." An invention producing a technical effect beyond the normal interactions between software and hardware is patent-eligible.

The 2024 EPO Guidelines clarified that:

- **AI and machine learning** are acceptable as technical means if applied to a technical purpose.
- **Mathematical methods** alone are not patentable but may be patentable when contributing to a technical character.
- **Training methods** for neural networks may be patentable when the trained network has technical application.

5.3.3 China

China has emerged as a leading jurisdiction for AI-related patent filings. Examination guidelines recognize AI, big data, and blockchain as technical fields when concrete technical solutions are disclosed.

5.3.4 India

Section 3(k) of the Indian Patents Act 1970 excludes "a computer program per se or algorithms" from patentability. However, the Indian Patent Office has granted patents to inventions involving computer programs with novel hardware integration or technical applications beyond mere data processing.

6. Practical Examples and Case Studies

6.1 Case Study 1: AI-Based Drug Discovery Platform

A research team at a CS-Biology convergence lab develops a deep learning model for predicting molecular binding affinities. They file:

- **India:** Provisional application filed on Day 1; complete specification filed within 12 months.
- **United States:** PCT national phase entry, claiming Paris priority from India.
- **Europe:** PCT regional phase entry at the EPO.
- **Japan:** PCT national phase entry at the JPO.

Strategic Justification: Commercial markets for AI-drug discovery include the US, EU, and Japan (large pharmaceutical markets). China is included given the growth of biotech in China. India is retained for potential local commercialization.

6.2 Case Study 2: Open-Source Software Library

A developer publishes a novel compression library on GitHub without filing a patent. Subsequently, a competitor incorporates the algorithm into a proprietary product. The developer seeks patent protection.

Analysis:

- In strict-novelty jurisdictions (Europe, India), the GitHub publication destroys novelty, and patent protection is unavailable absent a recognized grace period.
- In the US, a 12-month grace period under 35 USC §102(b)(1) may preserve patentability if the developer is the original inventor.
- Lesson: Researchers should evaluate IP protection strategy *before* public disclosure, even for open-source projects (where defensive patents may be valuable).

6.3 Case Study 3: PCT Application with Subsequent PPH Prosecution

A team files a PCT application claiming priority from a Korean provisional application. The ISR from the Korean Intellectual Property Office (KIPO) identifies limited prior art. The team enters the US national phase and requests PPH based on KIPO's positive examination, accelerating US prosecution.

Analysis: PPH programs typically reduce examination pendency by 30-50%, but require careful alignment of claims between jurisdictions.

7. Comparative Summary Table: Filing Routes

Filing Strategy	Best Suited For	Cost	Delay	Complexity
Paris Convention Direct National Filing	1-2 jurisdictions; urgent filing needs	Moderate to high per jurisdiction	Minimal	High (multiple procedures)
PCT Direct Filing (without priority claim)	First-time filers; no previous filing	High (includes PCT fees)	30 months to national phase	Moderate
PCT with Paris Convention Priority	Multi-jurisdiction; allows delay	Highest (PCT + national)	30 months from priority	Moderate
Regional Filing (e.g., EPO)	European coverage	Lower per country for multiple states	Similar to national	Moderate
ARIPO/OAPI Filing	African regional coverage	Cost-effective for African states	Variable	Moderate

8. Summary

This lecture has examined the multi-layered architecture of international cooperation in intellectual property, with particular attention to the procedures governing patent grants across jurisdictions. The principal points of synthesis are as follows:

- 1. Institutional Foundation:** WIPO serves as the central institutional hub for international IP cooperation, administering key treaties and providing global infrastructure (PATENTSCOPE, classification systems, arbitration services).
- 2. Treaty-Based Mechanisms:** The Paris Convention and the Patent Cooperation Treaty are the cornerstones of cross-border patent filing. The Paris Convention establishes the priority right framework; the PCT operationalizes unified international filing and search procedures.
- 3. Regional and Bilateral Frameworks:** Regional systems (EPC, ARIPO, OAPI, EAPO) and bilateral agreements (including PPH programs and FTA IP chapters) further harmonize and streamline patent procedures within defined groups of states.
- 4. Procedural Diversity:** Despite harmonization efforts, substantive and procedural variations persist across jurisdictions. These include differences in novelty standards, software patent eligibility, examination pendency, opposition mechanisms, and patent term calculations.
- 5. Strategic Imperatives for CS Researchers:** Researchers in computer science must integrate IP strategy into research planning, considering publication timing, commercial markets, jurisdictional patentability standards, and filing cost-benefit considerations.

The international IP system remains dynamic, with ongoing negotiations toward a Substantive Patent Law Treaty, evolving jurisprudence on AI and software patentability, and emerging regional agreements. Computer science researchers must remain informed of these developments to protect and leverage their innovations effectively.

9. Exam Preparation Tips

- 1. Memorize Key Treaties and Dates:** Know the foundational treaties (Paris Convention 1883, PCT 1970, EPC 1973) and their principal provisions. Examiners often test treaty-specific knowledge.
- 2. Distinguish Institutional Roles:** Be prepared to differentiate WIPO, WTO (TRIPS), and regional patent offices. Confusion between WIPO-administered treaties and TRIPS is common.
- 3. Understand PCT Phases:** The distinction between international phase and national phase is critical. Be able to explain the timing (12-month priority, 18-month publication, 30-month national phase entry) and procedural steps.
- 4. Apply Comparative Analysis:** When asked about jurisdictional differences, structure responses using comparative criteria: novelty standard, inventive step, software patentability, examination pendency, opposition procedures.
- 5. Link Concepts to CS Applications:** Examiners appreciate answers that connect IP mechanisms to computer science contexts (e.g., software patents, AI patentability, open-source IP considerations).
- 6. Use Precise Terminology:** Distinguish between "international application," "international patent," "national phase application," and "regional patent." Sloppy terminology suggests incomplete understanding.
- 7. Practice Case-Based Questions:** Be prepared to advise on filing strategies for hypothetical research scenarios, demonstrating strategic thinking rather than mere recitation of facts.

10. Practice Questions

Short Answer Questions

1. Explain the principle of territoriality in patent law and how it necessitates international cooperation mechanisms.
2. What is the right of priority under the Paris Convention, and what is its significance for patent applicants in computer science?
3. Distinguish between the international phase and the national phase of a PCT application.
4. Outline the principal functions of WIPO as they relate to patent applicants.

Long Answer / Essay Questions

5. Compare and contrast the patentability of computer-implemented inventions under the Alice/Mayo framework in the United States with the "technical character" requirement at the European Patent Office. Discuss the implications for AI-related patent applications.
6. A computer science research team at an Indian university develops a novel algorithm for federated learning and contemplates international patent protection. Outline a strategic filing plan, discussing the role of the Paris Convention priority right, the PCT international phase, and the national phase entry considerations. Justify your choices with reference to commercial markets, filing costs, and enforcement considerations.
7. Critically evaluate the role of bilateral and regional IP agreements in facilitating cross-border patent protection. Use specific examples (e.g., EPO, PPH programs, FTA IP chapters) to illustrate your arguments.
8. Discuss the challenges and opportunities that software and AI patentability pose to the international patent system. Examine how different jurisdictions have responded and the implications for global harmonization.

Applied / Case-Based Questions

9. *Scenario*: A graduate student publishes a paper on a novel data compression technique on arXiv, then six months later files a patent application in India, the US, and via the PCT designating the EU and Japan. Analyze the patentability prospects in each jurisdiction, considering novelty standards and grace periods.
10. *Scenario*: A startup files a PCT application on a novel recommendation algorithm. The ISR identifies moderate prior art, and the Written Opinion suggests the claims lack inventive step over a combination of three references. Recommend a strategy for response, including claim amendments, preliminary examination under PCT Chapter II, and national phase prosecution planning.

11. Glossary of Key Terms

- **Alice/Mayo Test**: A two-step framework established by US Supreme Court decisions (*Alice Corp. v. CLS Bank* and *Mayo Collaborative Services v. Prometheus*) for assessing patent eligibility of abstract ideas, including software and algorithms.
- **Budapest Treaty**: A WIPO-administered treaty (1977) for international recognition of microorganism deposits, relevant to biotech-computer science intersections.
- **COMVIK Approach**: An EPO jurisprudence principle (T 641/00) establishing the assessment of technical character in computer-implemented inventions.
- **European Patent Convention (EPC)**: The 1973 treaty establishing the European Patent Office and the European patent system.
- **International Preliminary Report on Patentability (IPRP)**: A non-binding assessment of patentability issued under PCT Chapter II.
- **International Search Report (ISR)**: A document identifying prior art relevant to a PCT application.
- **National Phase**: The post-PCT stage at which applications are processed in individual jurisdictions.
- **Paris Convention**: The 1883 multilateral treaty establishing foundational principles of international industrial property protection.
- **Patent Cooperation Treaty (PCT)**: The 1970 WIPO-administered treaty facilitating unified international patent filing procedures.
- **Patent Prosecution Highway (PPH)**: Bilateral or multilateral arrangements allowing accelerated examination based on positive findings in another jurisdiction.
- **TRIPS Agreement**: The 1994 WTO agreement establishing minimum IP standards among WTO member states (administered by WTO, not WIPO, but closely linked).
- **Unified Patent Court (UPC)**: A supranational court established to adjudicate European patent disputes (operational from 2023).

12. Suggested Readings

1. **WIPO Official Website:** www.wipo.int — Primary source for treaties, PCT procedures, PATENTSCOPE database, and policy documents.
 2. **World Intellectual Property Organization (2024):** *PCT Applicant's Guide*, latest edition — Definitive procedural reference for PCT filings.
 3. **European Patent Office (2024):** *Guidelines for Examination in the European Patent Office* — Essential for understanding EPO practice on computer-implemented inventions.
 4. **World Intellectual Property Organization (2019):** *WIPO Technology Trends 2019: Artificial Intelligence** — Authoritative study on AI patent landscapes.
 5. **Singh, A. (2016):** *Patent Law and Intellectual Property Rights in India*, LexisNexis — Comprehensive coverage of Indian patent practice.
 6. **Bodenhause, G.H.C. (1968):** *Guide to the Application of the Paris Convention for the Protection of Industrial Property* — Classical treatise on the Paris Convention (revised editions available).
 7. **UNCTAD-ICTSD Resource Book on TRIPS and Development** — Accessible resource on international IP standards.
 8. **Basheer, S., & Reddy, P. (Eds.) (2022):** *Intellectual Property Rights and Innovation in Computer Science*, Springer — Relevant contemporary perspectives on CS-related IP.
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Lecture Conclusion

International cooperation on intellectual property represents both an institutional achievement and an ongoing project. The mechanisms examined in this lecture—WIPO, the Paris Convention, the PCT, regional systems, and bilateral agreements—form an intricate web that enables researchers and inventors to navigate the territoriality principle. For computer science scholars, mastery of these mechanisms is essential, both for protecting their own innovations and for understanding the global IP landscape that increasingly shapes research, commercialization, and technology policy.

The procedural complexities across jurisdictions demand strategic thinking, careful timing, and informed professional advice. Yet the unifying principles—national treatment, priority rights, and international cooperation—provide the conceptual scaffolding that makes cross-border patent protection both feasible and rational.

End of Lecture 6 Notes